

The age-structured model required age-specific estimates of mean total length (TL), mean weight, survival, and vulnerability to harvest and catch to derive estimates of age-specific abundance (N_a). Age-specific mean TL was calculated using fixed-length subsamples (DeVries and Frie 1996) within a von Bertalanffy growth model (von Bertalanffy 1938). Mean weight (wt_a) was calculated for each age applying a length-weight equation parametrized for White Crappie using the mean TL at each age. To derive natural mortality estimates, several transformations were made using growth, longevity, and temperature metrics for crappie (e.g., Pauly, Hoenig, Jensen, Lorenzen; see Maceina and Sammons (2016) or Hilling et al. (2018) for further details on transformations). The mean of these transformations was roughly equivalent to estimates of the Brody coefficient (K) under each growth scenario, thus we simplified our estimate of natural mortality such that it was equivalent to the Brody coefficient. Model parameters can be found in Table 1.

Survivorship in the absence of fishing was calculated using the finite annual natural survival rate (S_0) as:

$$l_a = S_0 \times l_{a-1}. \quad (1)$$

Vulnerabilities for catch, harvest, and memorable length were calculated as:

$$V_a = \frac{1}{1 + e^{\frac{TL_a - TL_{50}}{SD}}}, \quad (2)$$

where V_a is the vulnerability at each age a , TL_a is mean TL at age a , TL_{50} is the TL at 50% vulnerability to capture, harvest, or reach memorable length, and SD is the standard deviation of TL_{50} . The TL_{50} parameter was set to 204, 254, and 305 mm for capture, harvest, and memorable length, respectively based on Miranda and Dorr (2000), a 254 mm (10-inch) minimum length limit (a commonly applied regulation across the U.S.), and Gabelhouse Jr. (1984). Values of SD

were used to specify the steepness of each vulnerability curve and set at 5% of the respective TL_{50} value (Table 1). Harvest and catch vulnerabilities were applied to calculate survivorship within the fished condition accounting for natural mortality, discard mortality, and fishing mortality as:

$$l_{fa} = l_{fa-1} \times S_0 \times (1 - U \times V_{harv}) \times (1 - (U_0 \times V_{cap} - U \times V_{harv}) \times D), \quad (3)$$

where l_{fa} is the survivorship in the fished condition, U is the finite annual exploitation rate which was set in each equation to the $U_{inflate}$ value, U_0 is the finite rate of capture by anglers, V_{harv} is the age-specific harvest vulnerability, V_{cap} is the age-specific capture vulnerability, and D is the discard mortality rate for White Crappie adapted from Muoneke (1992). The value of U_0 was set to U plus 10%, which allowed the model to portray some voluntary release of fish legal to harvest.

Incidence functions, per Walters and Martell (2004), were applied to account for the cumulative effects of fishing on the reproductive capacity of the population for the unfished (Φ_0) and fished (Φ_f) egg production per recruit as:

$$\Phi_0 = \sum_a f_a \times l_a \quad (4)$$

$$\Phi_f = \sum_a f_a \times l_{fa}, \quad (5)$$

where f_a represents age-specific fecundity. Although White Crappie generally exhibit a size of maturity at a TL of 180 mm (Bunnell et al. 2005; Meyer et al. 2022), this is a metric that can vary across crappie populations. Other age-structured modeling approaches assessing the overfishing potential for crappie have used fixed maturity sizes as high as 245 mm (e.g., Dotson et al. 2009), such that we added variability to size at maturity as this can greatly affect reference

points for overfishing. To that end, we set total length at maturity for each simulated year following a normal distribution with a mean of 200 mm and a SD of 20 mm. Weight at maturity was then calculated for each maturity length using the length-weight equation parametrized for White Crappie with a mean weight of maturity at 0.148 kg.

Age specific fecundity was estimated as a function of mean weight at age derived as the difference between mean weight at age and the weight at maturity. Fecundity was measured as zero where mean weight at age was less than the weight at maturity. While this measure of fecundity was not exact, the fecundity of crappie is strongly correlated to fish size (Bunnell et al. 2005) indicating that these estimates can serve as a suitable surrogate to calculate the potential for overfishing.

We used spawning potential ratio (SPR) to evaluate the extent that fishing mortality can reduce the reproductive output of a population. SPR is a common reference point used by fishery managers to assess the potential for recruitment overfishing (Goodyear 1993). SPR was calculated as:

$$\frac{\Phi_0}{\Phi_f} \quad (6)$$

and a static reference point of 30% was used as our overfishing metric as this metric has been applied to identify recruitment overfishing for other crappie populations (Dotson et al. 2009).

The variable R_{eq} (simulated recruitment) was calculated to predict the equilibrium number of age-1 recruits as

$$R_{eq} = R_0 \times Env, \quad (7)$$

where R_0 is the equilibrium number of age-1 recruits (set to 10,000 fish) and Env was a term simulating environmental stochasticity with a mean of 1 following a log-normal distribution at a coefficient of variation of 0.8 to mimic the high recruitment variability typically exhibited by crappie (e.g., Allen and Miranda 1998). The R_0 parameter was a fixed scaling parameter and did not influence model predictions. Our model assumed constant average recruitment and thus was a traditional yield-per-recruit analysis (Walters and Martell 2004).

Yield was calculated for each exploitation level as

$$yield = U \times \sum_a V_{harv} \times wt_a \times l_{fa}, \quad (8)$$

where V_{harv} is the harvest vulnerability for each age class calculated in Equation 5 and wt_a is the mean weight at each age. Yield per recruit (YPR) was calculated as yield divided by R_{eq} . As an additional metric to evaluate the population, an incidence function for the total number of memorable sized crappie in the fished population was calculated as:

$$\Phi_t = \sum_a V_{mem} \times l_{fa} \quad (9)$$

where V_{mem} is the age-specific memorable length vulnerability calculated in Equation 5. The proportion of memorable sized fish was calculated as the memorable population divided by total population.

References

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